BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 213 · Timed Paper 1 data and logic practice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which sort builds an ordered prefix by inserting a key?

Q2. How do you compensate for shifting a positive fractional mantissa left?

Q3. Simplify ¬(A ∨ B).

Q4. Find the decimal value of $0.1100_2 \times 2^3$.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

sign extension — Copy the sign bit into a new high-order position.

De Morgan — Swap AND/OR while negating components.

normalisation — Remove redundant leading mantissa sign bits.

mantissa — The fractional significant part.

exponent — Power-of-two scale.

New learning

Show positive first — Use exactly eight bits for +11, rather than padding after converting.

Invert each bit — 00001011 becomes 11110100.

Add one — 11110100 + 1 = 11110101 for −11.

Worked example

Model −6 in eight-bit two's complement.

1. Predict first — Write +6 as 00000110.

2. Check the method — Invert to 11111001.

3. Check the method — Add one to get 11111010, representing −6.

We Do · Practise with guidance

Find −9 in eight-bit two's complement.

`+9=00001001; invert; add 1.`

Predict first

Write 9 as 00001001.

Check the method

Invert to 11110110.

Add one to get 11110111, representing −9.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

(a) Represent −19 as an 8-bit two's-complement number. [4] (b) Perform an arithmetic right shift by 1 of your result. State its signed decimal interpretation. [4]

Question 2

(a) Use De Morgan's law to simplify ¬(A ∧ ¬B). [3] (b) Explain how an eight-cell group of ones simplifies a three-variable Karnaugh map. [5]

Question 3

A positive two's-complement fractional mantissa is 0.0011_2 with exponent 5. Normalise it, explaining each shift and why the value is unchanged.

Exit Ticket · Answer before checking

An arithmetic right shift of −19 by one place produces −9.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Insertion sort.

Q2. Decrease the exponent by the same number of places.

Q3. ¬A ∧ ¬B.

Q4. 0.75×8=6.

You Do pages · Independent questions

Q1. [8] (a) +19=00010011 [1], invert=11101100 [1], add 1=11101101 [2]. (b) Arithmetic right shift replicates leading 1 [1], gives 11110110 [2], interpreted as −10 [1]; arithmetic right shift of odd negative values rounds towards negative infinity.

Q2. [8] (a) ¬A ∨ ¬¬B [2], simplifying to ¬A ∨ B [1]. (b) The group covers all eight 3-variable combinations [2]; all three variables vary [1], leaving no literal [1], so output is constant 1 [1].

Q3. [8] $0.0011\times2^5 \rightarrow 0.0110\times2^4$ [2] $\rightarrow 0.1100\times2^3$ [2]; positive normalised mantissa begins 01 [1]; shifting left doubles mantissa and reducing exponent compensates [2]; decimal value 6 [1].

Exit Ticket · Model answer

False. It produces −10 in two's complement due to sign extension and rounding towards negative infinity.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding